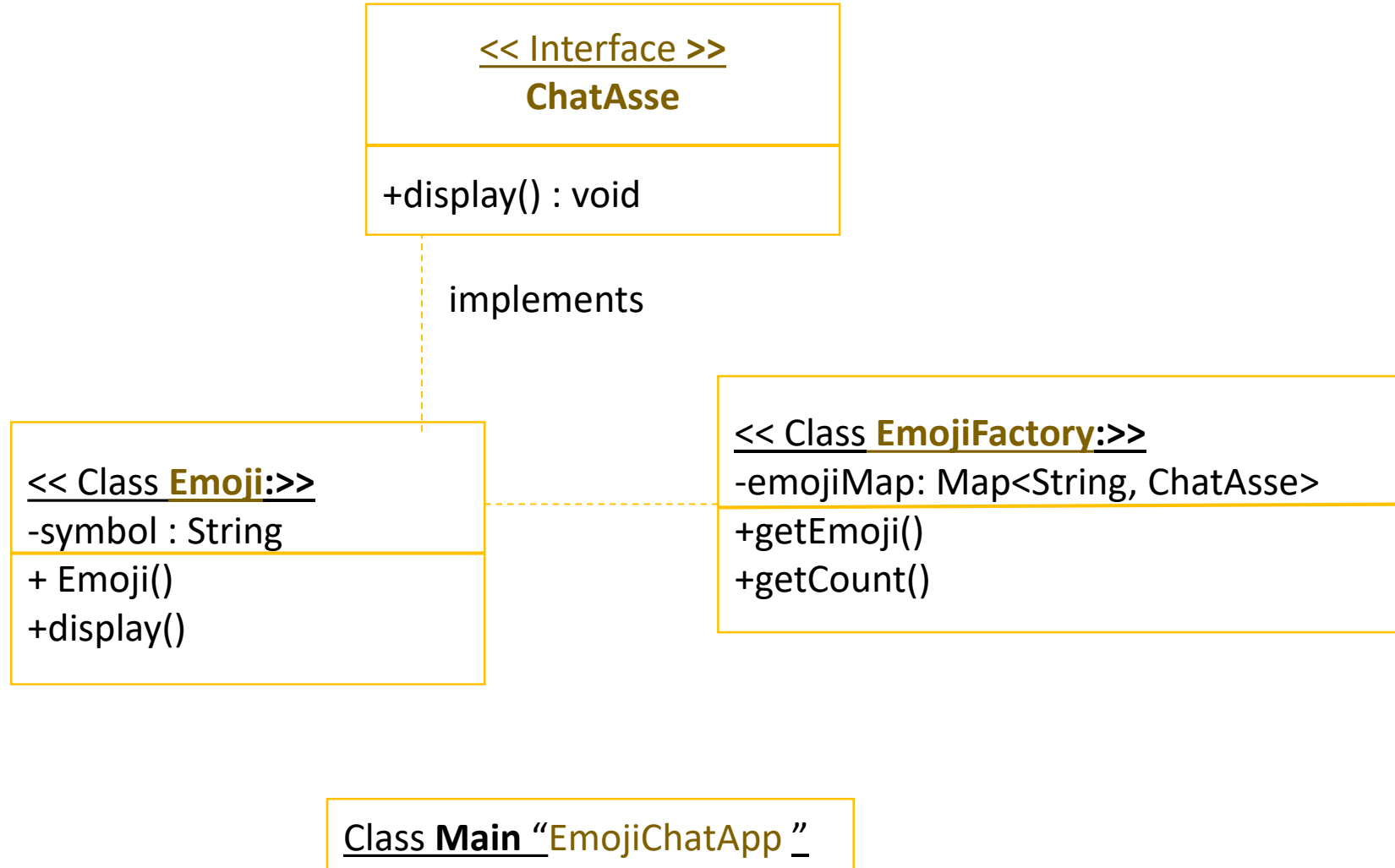




Package “emojichatapp”



Technical Documentation

Design Documentation

Project Title

Emoji Chat Application using Flyweight Design Pattern

1. Purpose of the System

The system simulates a simple chat application where users send messages with emojis.

Instead of creating a new emoji object every time a message is sent, the application reuses existing emoji objects from memory to reduce memory usage and improve performance.

2. Design Pattern Used

Flyweight Pattern

The Flyweight pattern is used to minimize memory consumption by sharing reusable objects instead of creating duplicate objects.

In this project:

- Emojis are shared objects.
- The factory stores already-created emojis.
- If the same emoji is requested again, the existing object is returned instead of creating a new one.

Example:

- 😊 is created once only.
 - ❤️ is created once only.
 - Both are reused multiple times.
-

3. Problem Solved

Without Flyweight Pattern:

- Every message creates a new emoji object.

- Memory usage increases unnecessarily.

With Flyweight Pattern:

- Repeated emojis share the same object.
- Fewer objects are stored in memory.
- Better performance and optimization.

4. Classes Description

4.1 ChatAsse (Interface)

Represents the common behavior for all chat assets.

Responsibilities

- Defines the `display()` method.

Methods

Method	Description
<code>display()</code>	Displays username, message, and emoji

4.2 Emoji Class

Concrete Flyweight object that stores the emoji symbol.

Attributes

Attribute	Type
<code>symbol</code>	String

Responsibilities

- Stores emoji symbol.
- Displays chat messages with emoji.

4.3 EmojiFactory Class

Factory responsible for creating and reusing emoji objects.

Attributes

Attribute	Type
<code>emojiMap</code>	<code>Map<String, ChatAsse></code>

Responsibilities

- Checks if emoji already exists.
- Creates new emoji only if needed.
- Returns shared emoji objects.

Methods

Method	Description
<code>getEmoji()</code>	Returns shared emoji object
<code>getCount()</code>	Returns total unique emoji objects

4.4 EmojiChatApp Class

Main class that runs the application.

Responsibilities

- Sends chat messages.
 - Requests emojis from factory.
 - Displays results.
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5. Advantages of the Design

- Reduces memory usage.
 - Prevents duplicate objects.
 - Improves application efficiency.
 - Easy to extend with new emojis.
-

6. Output Explanation

Although 4 messages are sent:

```
Total message sent in chat: 4
```

Only 2 actual emoji objects are created:



because the emojis are reused through the factory.

7. Conclusion

The Flyweight Pattern successfully optimized the chat application by sharing emoji objects instead of creating duplicates. This approach reduces memory consumption and demonstrates efficient object management in Java applications.